GOVERNMENT EXECUTIVE SUMMARY

Analysis in Global Supply Chain Operations

Delivery Performance, Delay Risk, and Logistics Efficiency

Prepared on: 07 November 2026

This executive summary presents the project's measured findings, performance indicators, operational challenges and recommended interventions. All quantitative findings are derived from the supplied project datasets.

Executive Overview

This report presents an evidence-based assessment of delivery performance and logistics efficiency using 180,519 project records. After excluding cancelled shipments and records with missing or invalid delivery durations, 172,765 eligible deliveries were assessed. The analysis identifies 98,977 delayed deliveries, 32,196 exactly on-time deliveries and 41,592 early deliveries. Overall SLA compliance, defined as delivery on or before the scheduled duration, is 42.71%. The report summarizes shipping-mode, regional, market and customer-segment performance and presents the five required project KPIs.

Supply Chain Problem

Delivery delays and inconsistent service performance can affect supply-chain reliability, customer experience, inventory planning and logistics costs. The project evaluates observed delivery durations against scheduled durations to identify where delivery performance is weaker. The analysis is descriptive: it identifies measured differences and operational exposure but does not establish the causes of individual delays.

Dataset Overview

The analysis uses the cleaned logistics dataset and the project's delivery-gap analysis dataset. The delivery-gap dataset contains 180,519 records and 39 columns. The cleaned dataset contains 180,519 records and 32 columns. The analysis uses actual shipping duration, scheduled shipping duration, delivery status, the existing binary late-delivery risk label, shipping mode, order region, market and customer segment.

Delivery Population

Population Measure	Records
Total Records	180,519
Eligible Deliveries	172,765
Cancelled Records	7,754
Invalid or Missing Duration Records	0
Delayed Deliveries	98,977
Exactly On-Time Deliveries	32,196
Early Deliveries	41,592

Delivery Performance Findings

Among 172,765 eligible deliveries, 98,977 (57.29%) were delayed, 32,196 (18.64%) arrived exactly on schedule, and 41,592 (24.07%) arrived early. SLA compliance was 42.71%. Average delay among delayed deliveries was 1.62 days, with a median of 1.00 days.

Delivery Classification Summary

Delivery Classification	Records	Percentage
Delayed	98,977	57.29
Exactly On-Time	32,196	18.64
Early	41,592	24.07
SLA Compliant (On-Time or Early)	73,788	42.71

Delay Risk Findings

The recorded Late Delivery Risk Ratio is 57.29%. The existing risk field is treated as an observed dataset label rather than a newly trained prediction. The delivery-gap analysis also shows that 57.29% of eligible records had a positive delivery gap. The highest observed delay was 4.00 days.

Shipping Mode Findings

The shipping mode with the highest calculated efficiency index is Standard Class. The mode-level results show differences in SLA compliance, late-delivery risk and delivery-gap performance. These results support evidence-based review of shipping-mode allocation.

Shipping Mode Comparison

Shipping Mode	Eligible_Deliveries	SLA_Compliance_Percent	Delayed_Percent	Late_Risk_Percent	Shipping_Mode_Efficiency_Index
Standard Class	103,153	60.23	39.77	39.77	68.18
Same Day	9,293	52.07	47.93	47.93	56.85
Second Class	33,806	20.17	79.83	79.83	16.14
First Class	26,513	0.00	100.00	100.00	9.97

Regional and Market Findings

The highest Regional Delay Index is observed in Central Asia. The market with the highest observed delayed-delivery percentage is Europe. These findings identify areas that warrant additional operational review.

Regional Performance

Order Region	Eligible_Deliveries	Delayed_Percent	SLA_Compliance_Percent	Regional_Delay_Index
Central Asia	542	56.46	43.54	100.00
Eastern Asia	6,973	56.72	43.28	98.55
South of USA	3,855	58.52	41.48	98.38
West of USA	7,595	56.79	43.21	97.86
Northern Europe	9,408	56.25	43.75	97.25
South Asia	7,455	58.35	41.65	97.14
Western Europe	25,867	58.53	41.47	97.08
West Africa	3,571	54.69	45.31	97.06
Eastern Europe	3,785	57.65	42.35	96.91
Caribbean	7,951	55.53	44.47	96.87
Canada	907	51.60	48.40	96.80
Oceania	9,733	56.32	43.68	96.47
South America	14,184	57.18	42.82	96.45
Central America	27,174	57.11	42.89	96.09
Central Africa	1,616	60.15	39.85	95.73
North Africa	3,086	57.10	42.90	95.42
West Asia	5,746	57.81	42.19	95.27
Southeast Asia	9,136	57.98	42.02	95.22
East Africa	1,781	58.17	41.83	95.05
East of USA	6,617	58.17	41.83	94.95
US Center	5,653	57.53	42.47	94.81
Southern Europe	9,030	56.80	43.20	92.92
Southern Africa	1,100	56.09	43.91	91.47

Market Performance

Market	Eligible_Deliveries	Delayed_Percent	SLA_Compliance_Percent
Europe	48,090	57.69	42.31
USCA	24,627	57.41	42.59
Pacific Asia	39,585	57.38	42.62
LATAM	49,309	56.87	43.13
Africa	11,154	56.84	43.16

Customer Segment Impact

Home Office has the highest delayed-delivery percentage among the analyzed customer segments. Segment-level results should be used to review service exposure without assuming that any segment represents a premium or enterprise category. The segment comparison also reports eligible delivery volume, average delivery gap, delayed-delivery percentage and SLA compliance.

Customer Segment Performance

Customer Segment	Eligible_Deliveries	Delayed_Percent	SLA_Compliance_Percent	Average_Delivery_Gap
Home Office	30,817	57.59	42.41	0.58
Consumer	89,420	57.31	42.69	0.57
Corporate	52,528	57.08	42.92	0.56

Key Performance Indicators

The following five KPIs are the required project indicators. The On-Time Delivery Rate follows the project's SLA-compliance definition: deliveries arriving on or before the scheduled duration.

Five Required Logistics KPIs

KPI	Value	Formula
On-Time Delivery Rate (%)	42.71	SLA-compliant eligible deliveries / eligible deliveries × 100
Average Delivery Delay (Days)	1.62	Mean positive delivery gap among delayed eligible deliveries
Late Delivery Risk Ratio (%)	57.29	Mean valid binary late-delivery risk label × 100
Shipping Mode Efficiency Index	37.78	Average of mode-level indices using 50% SLA, 30% inverse risk and 20% relative delay score
Regional Delay Index	100.00	Regional average positive delay / maximum regional average positive delay × 100

Identified Operational Challenges

- 98,977 eligible deliveries were delayed, representing 57.29% of the eligible population.
- Late-delivery risk exposure was 57.29% based on valid existing binary labels.
- Central Asia recorded the highest Regional Delay Index.
- Europe recorded the highest delayed-delivery percentage among markets.
- Home Office recorded the highest delayed-delivery percentage among customer segments.
- Shipping-mode performance varies across SLA compliance, late-delivery risk and delivery-gap measures.

Recommended Interventions

- Review delayed-delivery cases by shipping mode, region and market to identify operational causes using shipment-level records and supporting logistics information.
- Use the shipping-mode efficiency results as a screening tool when reviewing service allocation, while considering cost, capacity, route suitability and service commitments before changing allocations.

- Prioritize operational reviews in regions and markets with higher delay indicators. Validate local causes before implementing corrective actions.
- Review customer-segment service performance and assess whether service commitments, communication and exception handling are appropriate for each segment.
- Introduce routine monitoring of the five required KPIs and compare results across consistent reporting periods when reliable operational dates become available.
- Investigate late-delivery risk records and delivery exceptions to distinguish operational causes from the existing risk classification.
- Improve the completeness of shipment-level operational data, including genuine event dates, route details, carrier information and delay reasons.

Expected Operational Benefits

The following benefits are expected objectives of the recommended interventions. They are not measured savings or guaranteed outcomes.

- More consistent monitoring of delivery performance and SLA compliance.
- Earlier identification of areas with comparatively high delivery delays when timely operational data is available.
- Better-informed shipping-mode allocation reviews using measured efficiency indicators.
- More targeted regional, market and customer-segment performance reviews.
- Improved management reporting through consistent definitions and repeatable KPI calculations.
- Stronger evidence for future logistics improvement initiatives and operational investigations.

Data Limitations

- The dataset does not contain a genuine operational date field suitable for reliable chronological trend or seasonality analysis. Synthetic dates have not been created.
- The existing Late_delivery_risk field is a recorded binary label. It is not a newly trained predictive model or proof of future delay.
- The dataset does not provide a dedicated Order ID for treating customer identifiers as shipment identifiers. Customer IDs must not be used as order IDs.
- Customer latitude and longitude are not verified shipment-destination coordinates and are not used to create destination maps.
- The analysis excludes cancelled shipments and records with missing or negative delivery durations from delivery-performance denominators. Valid zero scheduled-duration Same Day records are retained.
- The analysis is descriptive and does not establish causality between shipping mode, geography, customer segment and delivery delays.
- The expected benefits are operational objectives, not quantified or guaranteed savings. Their realization requires implementation and subsequent measurement.

Conclusion

The analysis establishes a measurable baseline for delivery performance and logistics efficiency. SLA compliance is 42.71%, while 57.29% of eligible deliveries were delayed. Differences across shipping modes, regions, markets and customer segments provide practical areas for operational review. Government and supply-chain decision-makers can use the five project KPIs and supporting comparisons to prioritize investigations, strengthen monitoring and evaluate future improvements. Further causal and time-based analysis will require more complete operational data.

References

- APL Logistics source dataset: data/raw/APL_Logistics.xlsx.
- Project cleaned dataset: data/processed/cleaned_logistics_data.csv.
- Project delivery-gap dataset: data/processed/delivery_gap_analysis.csv.
- Project-generated KPI definitions and analysis outputs.
- Pandas documentation: https://pandas.pydata.org/docs/.
- NumPy documentation: https://numpy.org/doc/.